

Answer Sheet - ULMS Testing Assessment

ULMS Software Testing and Debugging Assessment

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Declaration of Authenticity

I, Johannes Mathopo, declare that this ULMS software testing and debugging assessment submission is my own work. I confirm that the test documentation, defect reports, debugging corrections and final testing report are aligned with the provided Umoja Learning Management System case study. I understand that copied or incomplete work may result in academic penalties.

Signature: Johannes Mathopo

1.1 Testing Objectives

System purpose: The Umoja Learning Management System (ULMS) WinForms application supports login authentication, student registration, course enrolment, marks capture, report generation and logout/navigation control.

Testing goals: verify that each module meets functional requirements, handles invalid input safely, calculates student averages correctly, prevents unauthorised access, and produces reports within an acceptable response time.

Scope of testing: login, dashboard navigation, student registration, course enrolment, marks capture, report generation, validation messages, basic security, usability, reliability and performance.

1.2 Testing Types Identification

Testing Type	ULMS Example	Method/Technique	Expected Outcome
Functional testing	Login validation	Positive and negative tests using valid and invalid credentials	Only valid admin/1234 credentials allow dashboard access
Input validation	Student registration and marks entry	Boundary value analysis and equivalence partitioning	Blank, non-numeric and out-of-range data rejected with clear messages
Integration testing	Dashboard to forms and course allocation	Navigation and workflow tests	Forms open correctly and return safely to dashboard
Performance testing	Report loading	Manual stopwatch/load timing over repeated runs	Report generation completes within 2 seconds
Usability testing	Navigation and clear buttons	Observation checklist	Labels, buttons and messages are understandable
Security testing	Authentication	Invalid login attempts and blank credentials	No access is granted unless username and password both match

1.3 Testing Environment Setup

Item	Specification
Hardware	Windows PC/laptop, minimum 8GB RAM, 64-bit processor
Operating System	Windows 10/11
Development Tool	Visual Studio 2022 or later with .NET Desktop Development

	workload
Framework	C# Windows Forms, .NET 8 Windows target
Test Data	Valid and invalid login credentials, valid/invalid student details, duplicate enrolment records, mark values 0-100, report filters
Evidence Tool	Windows Snipping Tool or Print Screen, ensuring taskbar clock/date is visible

Task 2: Structured Test Case Specification

Test Case ID	Feature	Input Data	Expected Output	Actual Output	Status
TC-001	Login validation	username=admin; password=1234	Dashboard opens and login success message displayed	Pass after correction	Pass
TC-002	Login validation	username=admin; password=wrong	Access denied message displayed; dashboard not opened	Fail before correction; Pass after correction	Pass
TC-003	Login validation	username=wrong; password=1234	Access denied message displayed; dashboard not opened	Fail before correction; Pass after correction	Pass
TC-004	Student registration	Valid ID, name, email, age 21, programme selected	Student saved successfully and summary displayed	Pass after correction	Pass
TC-005	Student registration	Blank student ID or name	Validation error; record not saved	Pass after correction	Pass
TC-006	Student registration	Age=abc	Validation error: age must be numeric	Fail before correction; Pass after correction	Pass
TC-007	Course enrolment	Valid student and course/semester selected	Enrolment completed and shown in output	Pass after correction	Pass
TC-008	Course enrolment duplicate	Same student, course and semester entered twice	Duplicate enrolment rejected	Fail before correction; Pass after correction	Pass
TC-009	Marks capture	Subject marks 78,65,80	Average is 74.33 and status PASS	Fail before correction; Pass after correction	Pass
TC-010	Marks capture boundary	Subject marks 49,50,51	Average is 50.00 and status PASS	Pass after correction	Pass
TC-011	Marks capture invalid	Subject1=-1 or 101	Validation error: marks must be 0- 100	Fail before correction; Pass after correction	Pass
TC-012	Report generation	Marks Report selected	Report displays correct marks and average 74.33, not 169	Fail before correction; Pass after correction	Pass
TC-013	Report generation performance	Student Summary Report selected	Report loads within 2 seconds	Fail before correction due to Thread.Sleep(4000); Pass after correction	Pass
TC-014	Dashboard navigation	Open each module from dashboard	Correct form opens; back/close returns to dashboard	Pass	Pass

Task 3: Functional Test Execution Log

Execution ID	Test Case IDs	Result	Evidence Required
EXE-001	TC-001 to TC-003	Login validation passed after correction	Full-screen screenshots of valid and invalid login attempts
EXE-002	TC-004 to TC-006	Student registration validation passed after correction	Screenshots of valid save and invalid age error
EXE-003	TC-007 to TC-008	Course enrolment duplicate control passed after correction	Screenshots of first enrolment and duplicate rejection
EXE-004	TC-009 to TC-011	Marks capture calculation and validation passed after correction	Screenshots showing average 74.33 and invalid mark rejection
EXE-005	TC-012 to TC-014	Report and navigation tests passed after correction	Screenshots showing report output and stopwatch/performance timing

Non-Functional Testing Results

Type	Method	Finding	Status
Performance	Generate each report type 3 times and record stopwatch duration	Before correction approximately 4 seconds due to Thread.Sleep(4000); after correction under 2 seconds	Pass after correction
Usability	Open dashboard modules, use clear/back buttons, check validation messages	Navigation and messages are clear; clear buttons reset fields and focus first field	Pass
Security	Attempt blank login and partial valid credentials	Access denied unless username and password both match	Pass after correction
Reliability	Repeat marks calculation and enrolment actions 5 times	No crashes after TryParse and validation added	Pass after correction

Task 5: Debugging Evidence - Summary

The following defects were identified in the supplied faulty modules and corrected in the included source files.

Debugging Evidence Table

Defect	Original Faulty Code	Issue	Corrected Code	Explanation
BUG-001 Login	if (username == "admin" password == "1234")	OR allowed partial credentials to login	if (username == "admin" && password == "1234")	Both username and password must be correct for authentication.
BUG-002 Average	record.Average = record.Subject1 + record.Subject2 + record.Subject3 / 3;	Division happened before addition	record.Average = (record.Subject1 + record.Subject2 + record.Subject3) / 3.0;	Parentheses ensure all three subjects are summed before division.
BUG-003 Marks input	Convert.ToDouble(txtSubject1.Text)	Blank or text input crashes application	double.TryParse(..., out double subject1)	TryParse allows validation error instead of unhandled exception.
BUG-004 Enrolment	No duplicate check before output	Same course can be enrolled repeatedly	Check existing enrolments with Any(...) before saving	Business rule prevents duplicate course registrations.

BUG-005 Reports	Thread.Sleep(4000);	Creates unacceptable delay	Removed Thread.Sleep and generated report immediately	Improves performance and meets report response requirement.
BUG-006 Age	Age = int.Parse(txtAge.Text)	Non-numeric age crashes application	int.TryParse(txtAge.Text, out int age)	Displays clear error instead of crashing.

Task 6: Defect Log / Bug Report

Bug ID	Module	Description	Severity	Priority	Status	Resolution
BUG-001	Login	Login used OR logic, allowing access when only username or password was correct	Critical	High	Fixed	Changed condition to username == admin AND password == 1234 and trimmed input
BUG-002	Marks Capture	Average calculation used incorrect operator precedence	High	High	Fixed	Changed to (Subject1 + Subject2 + Subject3) / 3.0
BUG-003	Marks Capture	Convert.ToDouble caused crash on non-numeric or blank input	High	High	Fixed	Replaced with double.TryParse and 0-100 range checks
BUG-004	Course Enrolment	Duplicate course enrolments were accepted	Medium	Medium	Fixed	Added in-memory duplicate validation using StudentId, CourseName and Semester
BUG-005	Reports	Report generation delayed by Thread.Sleep(4000)	Medium	Medium	Fixed	Removed artificial delay and used Stopwatch for timing
BUG-006	Student Registration	Age used int.Parse without validation	Medium	High	Fixed	Replaced with int.TryParse and sensible age range validation
BUG-007	Reports	Marks report displayed incorrect average 169	High	High	Fixed	Corrected report output to show 74.33 sample average

Executive Summary

Structured functional, non-functional and debugging tests were conducted for the ULMS WinForms application. The major risk was authentication bypass caused by incorrect OR logic in the login module. Additional defects were found in marks calculation, input validation, duplicate enrolment handling and report performance. Corrective actions were implemented and the application is considered ready for controlled submission testing after learner screenshots are added.

Testing Scope

Login, dashboard navigation, student registration, course enrolment, marks capture, report generation, input validation, security checks, usability checks, reliability checks and report performance.

Tools Used

Tool	Purpose
Visual Studio 2022+	Build, run and debug WinForms application
Manual test cases	Functional and non-functional test execution
Stopwatch/Date-time visible screen capture	Performance evidence
GitHub	Version control and repository submission
Windows Snipping Tool/Print Screen	Evidence screenshots

Functional Testing Results

Area	Result
Login	Passed after AND validation fix
Student Registration	Passed after age and required-field validation
Course Enrolment	Passed after duplicate validation
Marks Capture	Passed after formula and numeric validation corrections
Reports	Passed after performance and output corrections

Non-Functional Testing Results

Type	Method	Finding	Status
Performance	Generate each report type 3 times and record stopwatch duration	Before correction approximately 4 seconds due to Thread.Sleep(4000); after correction under 2 seconds	Pass after correction
Usability	Open dashboard modules, use clear/back buttons, check validation messages	Navigation and messages are clear; clear buttons reset fields and focus first field	Pass
Security	Attempt blank login and partial valid credentials	Access denied unless username and password both match	Pass after correction
Reliability	Repeat marks calculation and enrolment actions 5 times	No crashes after TryParse and validation added	Pass after correction

Defects Identified

Bug ID	Module	Summary	Status
BUG-001	Login	Login used OR logic, allowing access when only username or password was correct	Fixed
BUG-002	Marks Capture	Average calculation used incorrect operator precedence	Fixed
BUG-003	Marks Capture	Convert.ToDouble caused crash on non-numeric or blank input	Fixed
BUG-004	Course Enrolment	Duplicate course enrolments were accepted	Fixed
BUG-005	Reports	Report generation delayed by Thread.Sleep(4000)	Fixed

Debugging Corrections Implemented

The login condition was corrected from OR to AND. Marks calculation was corrected using parentheses and division by 3.0. Numeric parsing was hardened with TryParse. Age and marks ranges were validated. Course enrolment now blocks duplicate student/course/semester combinations. The report delay was removed and sample report output was corrected.

GitHub Link

Github